

Identifying Epileptic Animals from Resting-State GCaMP Imaging Alone

Subject-level classification from cortex-wide calcium dynamics, without electrophysiology and without capturing a seizure

A literature review of peer-reviewed and openly documented methods

Scope: widefield (mesoscale) cortex-wide imaging • resting-state / interictal data only • feature engineering and machine learning compared

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1. Problem Statement and Why This Is a Different Problem

1.1 The question being asked

- **Input:** cortex-wide widefield GCaMP imaging of a rodent at rest — no electrophysiology, no stimulus, no captured seizure.
- **Output:** a label for the *animal*: epileptic or not. The unit of prediction is the subject, not the frame or the event.
- **Why it matters:** it would remove the need for weeks of video-EEG to phenotype a colony, allow non-invasive longitudinal screening of epileptogenesis, and give a continuous severity readout for therapy trials rather than a seizure count.

1.2 Two versions of the problem, and they are not equally hard

- **Version A — interictal activity included.** Resting recordings are allowed to contain interictal epileptiform discharges (IEDs). Classification then partly reduces to IED detection, which is already solved (see the companion report). This version is much easier and much less interesting.
- **Version B — discharge-free background only.** Epochs containing IEDs are removed, and the classifier must work on ordinary background cortical dynamics. This is the scientifically meaningful claim: that the epileptic brain is measurably different even when it is behaving normally.
- This report treats **Version B as the target** and Version A as the fallback. Every study design recommendation assumes discharge-free epochs unless stated otherwise, and any published claim should state clearly which version it tested.

1.3 What 'resting state' has to mean in a head-fixed mouse

- Not merely 'no task'. Cortex-wide calcium signals are dominated by spontaneous movement: in a well-known analysis, single-trial cortical dynamics were shown to be dominated by richly varied movements rather than by task variables [23]. Uncontrolled movement will be the strongest feature in any naive classifier.
- The operational definition used in published epilepsy imaging is **quiet wakefulness**: awake, head-fixed, non-running epochs, with treadmill or wheel encoder and body video used to enforce it [1].
- Sleep is the other state confound. Widefield calcium data carry enough information to classify sleep state, and deep-learning sleep-state classifiers for exactly this data type have been built and validated [10] — which is both a warning and a usable tool.

Citation convention. Bracketed numbers refer to Section 10. Because this problem has not yet been solved in the published literature, the report distinguishes throughout between *demonstrated* findings and *transferable* methods.

2. Is There a Signal? Evidence That Resting Dynamics Differ

The whole enterprise rests on one empirical question: does the epileptic cortex look different between events? The evidence is real but uneven, and it is worth being precise about what has actually been shown.

2.1 Direct evidence in mesoscale calcium imaging

- **The strongest single result.** In *Kcnt1* gain-of-function mice imaged with cortex-wide GCaMP6s, interictal cortical events were classified by spatial profile and compared with wild-type. Event **rate was significantly altered in 5 of 10 event types** (increased in four), and every event type with an elevated rate had peak activity in areas susceptible to seizures and IEDs. Event **duration was reduced in 5 of 10 types** and increased in none [1].
- **A per-animal scalar biomarker already exists.** Applying SVD to an event-type × animal intensity matrix yielded a first right singular vector — effectively a single **global intensity scaling factor per animal** — and that scalar correlated with each mutant's epileptic activity burden, with a logistic fit relating the two [1]. This is the closest thing in the literature to a resting-state severity index from calcium imaging.
- **But the difference is quantitative, not qualitative.** The same study found that clustering pooled events produced clusters containing both genotypes, with highly correlated spatial profiles (median correlation 0.97), and **no novel event types** in mutants [1]. A classifier cannot look for a pathognomonic pattern; it must detect shifts in rate, duration and amplitude of otherwise normal activity.
- **Individual variability is large and may be the signal.** Wild-type animals showed consistent event intensities; mutants varied widely, some above and some below wild-type, consistently across sessions within an animal [1]. That is good news for per-animal features and bad news for group-mean thinking.
- **Progressive models show resting-state drift.** In glioma models, widefield imaging in awake mice tracked a reproducible progression from hyperexcitability to convulsive seizures [20], and documented desynchronisation between tumour and contralateral cortex alongside the onset of interictal events [19]. Cortical network metrics were dysregulated as a function of tumour stage and proximity [21].

2.2 Evidence from cellular-resolution and other preparations

- In Dravet (*Scn1a*^{+/-}) mice, mean activity of both putative principal cells and parvalbumin interneurons was **higher than wild-type during quiet wakefulness at baseline**, while the pre-ictal signature was a loss of interneuron synchronisation [17].
- **The honest counterweight:** other in vivo work in the same model reported *unaltered* network activity and interneuronal firing during spontaneous cortical dynamics, with the excitability deficit visible in slices but not in vivo at rest [18]. Resting-state separability is therefore model-dependent and should not be assumed.
- Whole-brain cellular imaging in *scn1lab* zebrafish found genotype-specific graph alterations, though most clearly under chemoconvulsant challenge rather than at baseline [22] — a hint that a mild provocation may be needed when baseline separation is weak.

2.3 Strong precedent from human electrophysiology

The human field has been asking exactly this question for a decade, on EEG and MEG rather than calcium, and its results set realistic expectations:

- Resting interictal MEG, using features from periods **free of any epileptiform activity**, discriminated healthy subjects from patients with idiopathic generalised or frontal focal epilepsy; relative power spectral density alone was sufficient for high accuracy [12].
- Directed functional connectivity estimated from **spike-free** high-density EEG epochs, fed to random-forest classifiers with leave-one-out validation, both diagnosed and lateralised temporal lobe epilepsy [13].
- A CNN applied to Granger-causality connectivity from just 50 s of resting surface EEG reached about **85% diagnostic accuracy** (F1 85%) in a 30-vs-30 cohort [14].
- **The necessary caution.** A careful clinical-validity study concluded that *univariate* resting-state EEG markers have limited validity for classifying seizure disorders [15], and a network-marker

study separating epilepsy from functional seizures reported a maximum balanced accuracy near 66% [16]. Single features will not carry this; multivariate models with honest validation are the minimum.

3. Designing the Recording

For subject-level classification the experimental design matters more than the algorithm, because the effects are small and every nuisance variable competes directly with the signal.

3.1 Matching and controls

- **Same reporter line for both groups, always.** Certain GCaMP6 genotypes, prominently *Emx1-Cre;Camk2a-tTA;Ai93*, produce spontaneous interictal-spike-like events above 0.1 Hz exceeding 10% $\Delta F/F$ across large cortical regions in otherwise normal animals [3]. In a seizure detector this is a nuisance; in a *classifier of epileptic animals* it is fatal if reporter genotype correlates with group.
- **Littermate controls**, same cohort, same surgery batch, same window type, interleaved imaging days. Window clarity and expression level both scale $\Delta F/F$ and are the most likely sources of a spurious classifier.
- **Blind the analyst** to genotype at the preprocessing and quality-control stage, since exclusion decisions are where unblinded bias enters.
- **Record enough animals.** The reference mesoscale epilepsy study used 6 mutants and 4 wild-types [1]. That is adequate for group-level statistics on thousands of events and inadequate for a classifier — see Section 6 for what happens to validation at that n.

3.2 State control and session structure

- Record wheel or treadmill position, body video and, ideally, pupil; restrict analysis to non-running, quiet-wakefulness epochs [1, 23].
- Either score sleep or exclude it. A validated deep-learning sleep classifier for widefield calcium data exists and can be used as a preprocessing gate [10].
- **Session length.** Delta-band functional connectivity maps in widefield GCaMP6 stabilise within windows as short as 30 s [6], but that is map stability, not feature reliability. For per-animal features, plan multiple 20–30 min sessions per animal across several days, following the design used in the reference study [1].
- **Repeat sessions are the test-retest measurement.** Report within-animal reliability of every candidate feature before reporting between-group differences; a feature whose session-to-session variance exceeds its group difference cannot classify an animal.
- **Anaesthesia.** Awake recording is strongly preferred: connectivity structure differs substantially between anaesthetised and awake states [6], and anaesthetic depth is a confound that varies with animal health. Where sedation is unavoidable, use a standardised protocol with physiological monitoring and objective depth measures [7].

3.3 Longitudinal design as a design advantage

- Chronic transcranial preparations allow the same animal to be imaged repeatedly over weeks [1, 7, 20], which converts a hard cross-sectional classification into an easier **within-animal change detection** problem.
- In acquired-epilepsy models (glioma, post-traumatic), pre-insult baselines are obtainable — making each animal its own control and allowing the question to be reframed as 'has this animal's resting cortex drifted?' rather than 'is this animal epileptic?' [19, 20, 24].

4. Preprocessing for Subtle Effects

The pipeline is broadly the one used for seizure detection, but the risk profile inverts: seizures are so large that preprocessing errors rarely flip a detection, whereas resting-state group differences are small enough that preprocessing choices can create or destroy them.

4.1 Core pipeline

- Dark-frame subtraction, channel de-interleaving, rigid motion correction [25].
- $\Delta F/F$ against a robust per-pixel baseline; the bottom-20th-percentile baseline used in the epilepsy literature is the safer default when discharge rate differs between groups [1].
- Hemodynamic correction — now the dominant methodological risk. Use a spatially detailed pixelwise regression, which outperformed simpler subtraction and Beer-Lambert approaches across three Cre lines [4, 5]. A group difference in vascular physiology would otherwise masquerade as a neural biomarker.
- Registration to the Allen CCF is **mandatory** here, not optional: cross-animal feature comparability is the entire premise. Automate with MesoNet and spot-check [26, 27].
- Denoising and compression by SVD or penalised matrix decomposition [25]. Fit the basis on pooled control and mutant data, never per group, or the representation itself becomes a group-informative feature.

4.2 Choices that will change the answer

- **Global signal regression.** Some widefield toolboxes regress the global signal by default. Given that a global intensity scaling factor is itself a candidate biomarker [1], GSR may remove the very effect being measured. Run the analysis both ways and report both.
- **Frequency bands.** Established widefield connectivity work separates infraslow (0.009–0.08 Hz), intermediate (0.08–0.4 Hz) and delta (0.4–4.0 Hz) bands, and connectivity structure differs markedly between them [6]. Band choice is a hypothesis, not a preprocessing detail.
- **Amplitude normalisation.** Z-scoring each animal removes expression and window confounds but also removes the global intensity biomarker. Keep both normalised and unnormalised feature sets and treat the difference as informative.
- **Discharge removal for Version B.** Detect and excise IEDs with the prominence-plus-duration criteria from the seizure-detection literature [1], excise a generous margin around each, and report how much data each group lost — asymmetric data loss is itself a confound.
- **Depth ambiguity.** Widefield fluorescence mixes superficial and deeper sources [28], so any 'cortical area' effect is a surface projection; describe findings accordingly.

5. Candidate Feature Families

Seven families, ordered roughly from best-evidenced to most speculative for this specific task. A realistic pipeline computes several and lets feature selection inside cross-validation decide.

5.1 Event-statistics features

- Detect all resting cortical events by peak prominence on a grid of ROIs, then characterise each by peak intensity, width at half max, spatial extent and atlas-area profile [1].
- Cluster events by spatial profile (hierarchical clustering on correlation distance, cluster number by silhouette analysis) and compute **per-animal rate, duration and intensity for each event type** [1].
- Reduce to a per-animal scalar by SVD across the event-type $\times$ animal matrix [1].
- **Status:** demonstrated to separate genotypes and to predict epileptic burden. This is the family to build first.

5.2 Functional connectivity features

- Seed-pixel correlation maps, atlas-area-by-area correlation matrices, and homotopic (interhemispheric mirror) correlation, all standard in widefield calcium work [6, 7].
- Graph metrics on the resulting matrix: node degree, clustering coefficient, modularity, hubness, characteristic path length.
- Multivariate rather than pairwise connectivity approaches have been developed specifically for widefield calcium and may be more sensitive than seed-based correlation [9].
- **Status:** demonstrated to detect disease states in other models (stroke, glioma, TBI) [7, 19, 24] and the dominant feature family in the human epilepsy-diagnosis literature [13, 14]. Not yet published for rodent epilepsy classification.

5.3 Spectral and temporal features

- Relative power spectral density per atlas area — notable because in interictal MEG this alone was sufficient for high classification accuracy [12].
- Autocorrelation time constants, signal variance, and the slope of the aperiodic spectrum as an indirect index of excitation-inhibition balance.
- **Caveat specific to calcium:** the indicator's kinetics low-pass the signal, and residual heartbeat and hemodynamic components occupy specific bands [4, 5]. Spectral features are therefore partly measurements of the preparation, and must be checked against GFP or wild-type controls imaged the same way.

5.4 Spatiotemporal dynamics and wave features

- Cortical activity cycles between distinct macroscale functional modules in a frequency-dependent way in normal mice [29], giving a natural set of state-occupancy and transition-rate features.
- Optical flow and finite-time Lyapunov exponent analysis (FLOW portraits) summarise where activity initiates and terminates and how it spreads, capturing coherent structures that modal decomposition misses [30].
- Propagation direction and speed of delta-band travelling waves differ between brain states [6] and are plausible correlates of altered excitability.
- **Status:** methodologically mature, untested as an epilepsy biomarker.

5.5 Criticality and avalanche features

- Healthy cortex is thought to operate near a critical point; deviation from criticality has been proposed as a general biomarker of excitation-inhibition imbalance.
- Directly relevant precedent: using MEG, deviations from critical dynamics were quantified in the **interictal** activity of epilepsy patients, with avalanche metrics providing a quantitative description of interictal brain activity [11].
- Usable metrics: avalanche size and duration exponents, branching ratio, shape collapse error, and the deviation-from-criticality coefficient, which are more discriminating than power-law fitting alone [31].
- **Serious caveat:** avalanche statistics are strongly biased by spatial subsampling and by how avalanches are thresholded [32]. Widefield pixels are not neurons; treat these as *relative*, *within-study* measures and never compare exponents across papers.

5.6 Massive unbiased time-series feature extraction

- Rather than hand-picking, compute thousands of generic time-series features per ROI with an automated phenotyping framework and let regularised selection find the discriminative ones [33].
- Attractive when there is no strong prior about which property differs, which is the current situation. It also converts the problem into a well-understood tabular classification task.
- Requires very strict nested cross-validation, since the feature-to-sample ratio becomes extreme.

5.7 Learned representations

- An LSTM autoencoder has been applied to widefield calcium data to learn spatiotemporal latent embeddings, from which functional networks were then estimated by regression, with spatial similarity to seed-based maps [8].
- Attention-based CNN-BiLSTM architectures have been validated on this exact data type for sleep-state classification [10], demonstrating that supervised deep models can learn subject-relevant state information from cortex-wide calcium frames.
- **Recommended use:** unsupervised embedding first, then a simple classifier on the embedding. With the animal counts available, end-to-end supervised deep learning on raw frames will memorise animals rather than learn pathology.

6. Classification at the Animal Level

6.1 The unit-of-analysis trap

This is the single most common way studies of this kind fail. Sessions and frames from the same animal are not independent samples. A model trained on frames and validated on held-out frames will report near-perfect accuracy that reflects animal identity, expression level and window quality, not epilepsy.

- **Split by animal, always.** Leave-one-animal-out or leave-one-pair-out nested cross-validation, with feature selection and hyperparameter tuning performed *inside* the training fold.
- Aggregate session-level predictions into one animal-level prediction (mean predicted probability, or majority vote) before scoring.
- Weight animals equally regardless of how many sessions they contributed, following the session-weighting practice already used in the reference study [1].

6.2 Model choice under small n

- With roughly 10–20 animals, prefer **regularised logistic regression or a random forest on a small, pre-registered feature set** over flexible models. The human spike-free EEG literature reached useful accuracy with random forests on connectivity features and leave-one-out validation [13].
- **Normative modelling / anomaly detection is often the better framing:** fit a distribution to control animals only, then score each test animal's deviation (Mahalanobis distance, one-class SVM, or an autoencoder reconstruction error). This avoids needing balanced groups and yields a continuous severity score rather than a binary label — which matches the finding that epileptic burden varies continuously across mutants [1].
- **Mixed-effects models are a legitimate alternative to ML entirely.** With ten animals, a hierarchical model with animal as a random effect, reporting effect size and confidence intervals, is often more informative and more defensible than a classifier.

6.3 Establishing chance level honestly

- With $n = 10$, a single held-out animal is 10% of the sample; accuracy estimates are extremely coarse and confidence intervals are wide. Report the interval, not the point estimate.
- Use **permutation testing**: shuffle animal labels, rerun the entire pipeline including feature selection, and build an empirical null. Anything less overstates significance.
- Follow the shuffle-based approach already established in this literature for group differences in event rate and duration, with correction across event types [1].

7. What the Classifier Might Actually Be Detecting

Before claiming an epilepsy biomarker, each of these alternatives has to be excluded explicitly. Most are testable with data already collected.

Alternative explanation	How it enters	How to test or control it
Reporter genotype	Certain GCaMP6 lines generate spontaneous epileptiform events independent of disease [3].	Identical reporter in both groups; verify the line is not on the affected list; image reporter-only controls.
Expression level and window clarity	Both scale $\Delta F/F$ amplitude, and both can correlate with genotype or surgery batch.	Report F_0 maps per animal; repeat analysis on z-scored data; interleave surgeries.
Behavioural state	Movement dominates cortex-wide signals [23]; sleep architecture may differ in epileptic animals.	Enforce quiet wakefulness; score sleep with a validated classifier [10]; test whether the classifier survives state matching.
Hemodynamics	Disease alters neurovascular coupling, and correction models are imperfect [4, 5, 19].	Compare correction methods; check that the effect is not present in the reflectance channel alone.
Residual undetected discharges	Sub-threshold IEDs left in 'discharge-free' epochs re-introduce Version A through the back door.	Sweep the detection threshold downward and show the effect is stable; excise generous margins.
The mutation, not the epilepsy	A gene variant may alter cortical dynamics whether or not that animal ever seizes.	The decisive test: within mutants, does the score track seizure burden? Two of six mutants in the reference study never seized [1].
Rig, batch and date	Groups imaged in different weeks or on different setups.	Record acquisition metadata as covariates; attempt to classify batch from the same features as a negative control.

The sixth row deserves emphasis. 'Epileptic animal' and 'animal carrying an epilepsy gene variant' are different labels, and only the first is clinically meaningful. The reference study found seizures in only four of six mutants, and the two without seizures were also the two with the lowest IED frequency [1] — so a well-designed study can, and should, test whether the resting-state score separates seizing from non-seizing mutants.

8. A Recommended Study and Analysis Plan

8.1 Phase 1 — Feasibility on existing data

- Take an existing chronic widefield dataset with video-EEG-confirmed phenotype. Compute the event-statistics feature set [1] on quiet-wakefulness, discharge-free epochs.
- Report **within-animal test-retest reliability** of every feature first. Discard features whose session-to-session variability exceeds the observed group difference.
- Report effect sizes with confidence intervals, not p-values alone, and do not fit a classifier yet.

8.2 Phase 2 — Feature expansion and normative modelling

- Add connectivity features (atlas-area matrices, homotopic correlation, graph metrics) in three frequency bands [6, 9], spectral features [12], and criticality metrics [11, 31].
- Fit a normative model on control animals only; score each animal's multivariate deviation.
- Test the two decisive relationships: does the score correlate with seizure burden within mutants, and does it separate seizing from non-seizing mutants?

8.3 Phase 3 — Classification with honest validation

- Regularised logistic regression or random forest, leave-one-animal-out nested CV, feature selection inside the fold, permutation null across the whole pipeline.

- Run the full confound battery in Section 7, including the batch-classification negative control.
- Report balanced accuracy with confidence intervals, per-animal predicted probabilities, and calibration — and follow TRIPOD-style reporting for prediction models [34].

8.4 Phase 4 — Generalisation, which is the real test

- Validate on a **held-out cohort** imaged later, ideally on a second rig.
- Validate across a **second epilepsy model** with different aetiology; a feature set that transfers from a genetic model to an acquired one (glioma or post-traumatic) [19, 20, 24] is measuring hyperexcitability rather than genotype.
- Only then consider learned representations [8, 10], using the classical model as the benchmark to beat.

9. Open Questions and Honest Limitations

- **No one has published this result.** As of this review there is no demonstrated subject-level classifier of epileptic animals from resting-state calcium imaging. The ingredients exist; the experiment does not.
- **Effect size is unknown and model-dependent.** The evidence ranges from clear interictal differences in *Kcnt1* mice [1] to reports of unaltered spontaneous cortical dynamics in *Scn1a*^{+/-} mice in vivo [18]. A negative result in one model would not generalise, and neither would a positive one.
- **Animal numbers are the binding constraint.** Typical cohorts of 4–10 per group [1] support group statistics but barely support classification. Multi-laboratory pooling, or a shared dataset, is probably a prerequisite — and none exists.
- **Human results set a ceiling expectation.** Balanced accuracies around 66–85% are what careful spike-free resting classification achieves in people [14, 16], and univariate markers were found to have limited clinical validity [15]. Rodent studies reporting 95%+ from ten animals should be assumed to have a validation leak until proven otherwise.
- **Timing within epileptogenesis is unaddressed.** Whether a resting-state signature appears before, with, or only after the first spontaneous seizure is the most valuable open question, and longitudinal glioma and post-traumatic models are the natural preparations in which to answer it [19, 20, 24].
- **Anti-seizure medication is an unexamined confound.** Any translational version of this assay will encounter treated animals, and drug effects on resting cortical dynamics have not been characterised at this scale.
- **A mild provocation may be necessary.** Where baseline separation is weak, genotype-specific network differences have emerged more clearly under chemoconvulsant challenge [22]. A sub-threshold challenge protocol would no longer be purely resting state, but may be the pragmatic compromise.

10. References

All sources were consulted online. Where an author list could not be verified from the accessed record, the entry is listed by title.

- [1] Tobin, W.F. & Weston, M.C. (2023). Distinct features of interictal activity predict seizure localization and burden in a mouse model of childhood epilepsy. *Journal of Neuroscience*, 43(27), 5076–5091. doi:10.1523/JNEUROSCI.2205-22.2023 — <https://www.jneurosci.org/content/43/27/5076>
- [2] Tobin, W.F. & Weston, M.C. (2025). Mesoscale Ca⁺⁺ imaging of seizures and interictal activity in models of chronic epilepsy. In: *Awake Behaving Mesoscopic Brain Imaging*, Neuromethods vol. 214, pp. 233–254. Humana, New York. doi:10.1007/978-1-0716-4120-0_10

- [3] Steinmetz, N.A., Buetfering, C., Lecoq, J., ... Carandini, M. & Harris, K.D. (2017). Aberrant cortical activity in multiple GCaMP6-expressing transgenic mouse lines. *eNeuro*, 4(5), ENEURO.0207-17.2017. doi:10.1523/ENEURO.0207-17.2017
- [4] Valley, M.T., Moore, M.G., Zhuang, J., Mesa, N., Castelli, D., Sullivan, D., Reimers, M. & Waters, J. (2020). Separation of hemodynamic signals from GCaMP fluorescence measured with wide-field imaging. *Journal of Neurophysiology*, 123(1), 356–366. doi:10.1152/jn.00304.2019
- [5] Ma, Y., Shaik, M.A., Kim, S.H., Kozberg, M.G., Thibodeaux, D.N., Zhao, H.T., Yu, H. & Hillman, E.M.C. (2016). Wide-field optical mapping of neural activity and brain haemodynamics: considerations and novel approaches. *Philosophical Transactions of the Royal Society B*, 371(1705), 20150360. doi:10.1098/rstb.2015.0360
- [6] Wright, P.W., Brier, L.M., Bauer, A.Q., Baxter, G.A., Kraft, A.W., Reisman, M.D., Bice, A.R., Snyder, A.Z., Lee, J.-M. & Culver, J.P. (2017). Functional connectivity structure of cortical calcium dynamics in anesthetized and awake mice. *PLOS ONE*, 12(10), e0185759. doi:10.1371/journal.pone.0185759
- [7] Cramer, J.V., Gesierich, B., Roth, S., Dichgans, M., Düring, M. & Liesz, A. (2019). In vivo widefield calcium imaging of the mouse cortex for analysis of network connectivity in health and brain disease. *NeuroImage*, 199, 570–584. doi:10.1016/j.neuroimage.2019.06.014
- [8] Zhang, X., Landsness, E.C., Brier, L.M., Chen, W., Tang, M.J., Miao, H., Lee, J.-M., Anastasio, M.A. & Culver, J.P. (2024). Identifying functional brain networks of spatiotemporal wide-field calcium imaging data via a long short-term memory autoencoder. arXiv:2405.19685. <https://arxiv.org/abs/2405.19685>
- [9] Brier, L.M., Zhang, X., Bice, A.R., Gaines, S.H., Landsness, E.C., Lee, J.-M., Anastasio, M.A. & Culver, J.P. (2022). A multivariate functional connectivity approach to mapping brain networks and imputing neural activity in mice. *Cerebral Cortex*, 32.
- [10] Attention-based CNN-BiLSTM for sleep state classification of spatiotemporal wide-field calcium imaging data; and Validation of deep learning-based sleep state classification. Zhang, X., Landsness, E.C., Culver, J.P., Lee, J.-M. & Anastasio, M.A. (Proc. SPIE; microPublication Biology, 2022). <https://anastasio.bioengineering.illinois.edu/research/machine-learning-in-neuroimaging/>
- [11] Arviv, O., Goldstein, A. & Shriki, O. (2016). Deviations from critical dynamics in interictal epileptiform activity. *Journal of Neuroscience*, 36(48), 12276. <https://www.jneurosci.org/content/36/48/12276>
- [12] Automated detection of epileptic biomarkers in resting-state interictal MEG data. *Frontiers in Neuroinformatics* (2017), 11, 43. doi:10.3389/fninf.2017.00043
- [13] Automated diagnosis of temporal lobe epilepsy in the absence of interictal spikes (directed functional connectivity, random forest, leave-one-out validation). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5842753/>
- [14] Hybrid machine learning method for a connectivity-based epilepsy diagnosis with resting-state EEG. *Medical & Biological Engineering & Computing* (2022). doi:10.1007/s11517-022-02560-w
- [15] Limited clinical validity of univariate resting-state EEG markers for classifying seizure disorders. *Brain Communications* (2023), 5(6), fcad330. <https://academic.oup.com/braincomms/article/5/6/fcad330/7456007>
- [16] Resting-state EEG network markers differentiate people with epilepsy and functional seizures (preprint). medRxiv (2026). <https://www.medrxiv.org/content/10.64898/2026.04.14.26350505>
- [17] Interneuron desynchronization precedes seizures in a mouse model of Dravet syndrome. *Journal of Neuroscience* (2020), 40(13), 2764. <https://www.jneurosci.org/content/40/13/2764>
- [18] Hull, J.M. & Isom, L.L. (2016). Expecting the unexpected: lack of in vivo network defects in an *Scn1a* model of Dravet syndrome. *Epilepsy Currents*, 16(6), 408. doi:10.5698/1535-7511-16.6.408
- [19] Montgomery, M.K., Kim, S.H., Dovas, A., et al. (2020). Glioma-induced alterations in neuronal activity and neurovascular coupling during disease progression. *Cell Reports*, 31, 107500. doi:10.1016/j.celrep.2020.03.064
- [20] Hatcher, A., Yu, K., Meyer, J., Aiba, I., Deneen, B. & Noebels, J.L. (2020). Pathogenesis of peritumoral hyperexcitability in an immunocompetent CRISPR-based glioblastoma model. *Journal of Clinical Investigation*. <https://www.jci.org/articles/view/133316>
- [21] Glioblastoma disrupts cortical network activity at multiple spatial and temporal scales. *Nature Communications* (2024), 15. doi:10.1038/s41467-024-48757-5
- [22] Whole-brain cellular-resolution functional network properties of seizure susceptibility (*scn1lab* zebrafish; graph analyses of pairwise correlations). bioRxiv (2026). <https://www.biorxiv.org/content/10.64898/2026.04.04.715761>

- [23] Musall, S., Kaufman, M.T., Juavinett, A.L., Gluf, S. & Churchland, A.K. (2019). Single-trial neural dynamics are dominated by richly varied movements. *Nature Neuroscience*, 22(10), 1677–1686. doi:10.1038/s41593-019-0502-4
- [24] Traumatic brain injury disrupts state-dependent functional cortical connectivity in a mouse model (widefield mesoscale imaging with simultaneous ECoG). bioRxiv (2023). <https://www.biorxiv.org/content/10.1101/2023.10.06.560776>
- [25] *wfield*: tools to process widefield data (motion correction, penalised matrix decomposition, hemodynamic correction, LocaNMF). <https://github.com/churchlandlab/wfield>
- [26] Xiao, D., Forys, B.J., Vanni, M.P. & Murphy, T.H. (2021). MesoNet allows automated scaling and segmentation of mouse mesoscale cortical maps using machine learning. *Nature Communications*, 12(1), 5992. doi:10.1038/s41467-021-26255-2
- [27] Wang, Q., Ding, S.-L., Li, Y., et al. (2020). The Allen Mouse Brain Common Coordinate Framework: a 3D reference atlas. *Cell*, 181(4), 936–953.e20. doi:10.1016/j.cell.2020.04.007
- [28] Waters, J. (2020). Sources of widefield fluorescence from the brain. *eLife*, 9, e59841. doi:10.7554/eLife.59841
- [29] Vanni, M.P., Chan, A.W., Balbi, M., Silasi, G. & Murphy, T.H. (2017). Mesoscale mapping of mouse cortex reveals frequency-dependent cycling between distinct macroscale functional modules. *Journal of Neuroscience*, 37(31), 7513–7533. doi:10.1523/JNEUROSCI.3560-16.2017
- [30] Linden, N.J., Tabuena, D.R., Steinmetz, N.A., Moody, W.J., Brunton, S.L. & Brunton, B.W. (2021). Go with the FLOW: visualizing spatiotemporal dynamics in optical widefield calcium imaging. *Journal of the Royal Society Interface*, 18(181), 20210523. doi:10.1098/rsif.2021.0523
- [31] Neural networks are tuned near criticality during a cognitive task and distanced from criticality in a psychopharmacological model (branching ratio, deviation-from-criticality coefficient, shape collapse; calcium imaging). bioRxiv (2024). <https://www.biorxiv.org/content/10.1101/2023.08.16.553626v2>
- [32] Neto, J.P., Spitzner, F.P. & Priesemann, V. Sampling effects and measurement overlap can bias the inference of neuronal avalanches. arXiv:1910.09984. <https://arxiv.org/pdf/1910.09984>
- [33] Fulcher, B.D. & Jones, N.S. (2017). *hctsa*: a computational framework for automated time-series phenotyping using massive feature extraction. *Cell Systems*, 5(5), 527–531. doi:10.1016/j.cels.2017.10.001
- [34] Collins, G.S., Reitsma, J.B., Altman, D.G. & Moons, K.G.M. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ*, 350, g7594. doi:10.1136/bmj.g7594
- [35] Shore, A.N., Colombo, S., Tobin, W.F., et al. & Weston, M.C. (2020). Reduced GABAergic neuron excitability, altered synaptic connectivity, and seizures in a KCNT1 gain-of-function mouse model of childhood epilepsy. *Cell Reports*, 33, 108303. doi:10.1016/j.celrep.2020.108303
- [36] Ren, C. & Komiyama, T. (2021). Characterizing cortex-wide dynamics with wide-field calcium imaging. *Journal of Neuroscience*, 41(19), 4160–4168. doi:10.1523/JNEUROSCI.3003-20.2021